

# Deepak Gopinath

<https://dpacgopinath.github.io>

[LinkedIn](#)

[Google Scholar](#)

[dpacgopinath@gmail.com](mailto:dpacgopinath@gmail.com)

## EDUCATION

---

### Carnegie Mellon University

M.S., Language Technologies Institute, School of Computer Science

Pittsburgh, PA

Aug 2015 – Dec 2016

### Birla Institute of Technology and Science, Pilani

B.S. in Computer Science

Pilani, India

2010 – 2014

## WORK EXPERIENCE

---

### Thinking Machines Lab

*Member of Technical Staff*

New York City, NY

Oct 2025 – current

- [Thinking Machines Lab](#) is an artificial intelligence research and product company aiming to make AI systems more widely understood, customizable and generally capable.

### Apple Inc

*Machine Learning Researcher, Foundation Models*

New York City, NY

March 2023 – Oct 2025

- Large Language Model (LLM) research as part of the core model training team for Foundation Models (FM) that power Apple Intelligence and FM Developer API [\[blog\]](#)[\[technical report\]](#)
- Leading efforts on pre-training, post-training and evaluation of language models, with a focus on web scale data, pretraining recipes and multilinguality. Heading multiple work streams that collectively engage 25+ contributors
- Skills: Python, Jax, Apache Beam on Dataflow, AXLearn/Ajax LLM training frameworks

### Meta AI (Facebook AI Research)

*Research Engineer, AI for Creativity*

Menlo Park, CA; Pittsburgh, PA

July 2019 – March 2023

- Lead Engineer for AI for Creativity supporting exploratory research in animating human characters with sequence models and reinforcement learning (RL). Developed and open-sourced the [fairmotion](#) and [ScaDiver](#) libraries to enable human motion research [\[11\]](#)
- Initiated and led cross-functional projects to apply research to production. Established collaborations with Habitat for Embodied AI research, Horizon Worlds VR product, Sanzaru game studio, FAIR Paris/Menlo Park, and The Movement Lab at Stanford University
- Published research at ACM SIGGRAPH [\[6\]](#) [\[9\]](#) [\[10\]](#), NeurIPS [\[8\]](#), and SIGGRAPH Asia [\[7\]](#) (under review)
- Tech lead for multiple projects involving 20+ engineers in total at Meta AI. Mentored 9 Research and SWE interns, supported 20+ new team members and mentored 5 junior engineers in technical mentorship programs
- Skills: Python, PyTorch, TorchServe; RLlib for distributed RL training; PyBullet and Isaac Gym physics simulators; OpenGL, Blender, Unity for character/scene building, animation and rendering.

*Research Engineer, Language and Translation Technologies*

Jan 2017 – July 2019

- Co-developed and shipped the first neural machine translation (NMT) system to power translations for all user content on Facebook and Instagram (7B+ translations per day) [\[blog\]](#)[\[press\]](#)
- Tech lead for a team that modeled and deployed the first multilingual NMT systems to production [\[blog\]](#) [\[press\]](#)
- Planned and managed “engineering excellence” projects that involved architecting optimized workflows for data processing, model training, evaluation and packaging, transitioning models to PyTorch, addressing reproducibility, determinism and safety in translations.
- Skills: Python, PyTorch; ONNX, C++ for production code; Hive, Spark for data processing

*Software Engineering Intern, Language Technology*

May 2016 – Aug 2016

- Developed neural network language models on the (newly) open sourced deep learning framework Caffe2.0

## LinkedIn

Software Engineer, Content Search

Bangalore, India  
July 2014 – Aug 2015

- Developed the prefix query tagger library to predict autocompletions and tags for an incomplete query
- Co-implemented and deployed federated search and auto-complete search for several verticals on linkedin.com
- Actively involved in recruitment and onboarding of university grad hires and co-mentored 3 SWE interns
- Skills: Java, Lucene, Pig, Flex, Play Framework with JQuery and Bootstrap, JavaScript

## Microsoft Research

Research Developer Intern, Applied Sciences

Bangalore, India  
Jan 2014 – July 2014

- Fraud detection in Azure services and sentiment analysis of Azure users in social media
- Launched the Indian Elections Game web app, an interactive prediction market for the public to make predictions on the 2014 Indian Elections [\[blog\]](#)

## Amazon

Machine Learning Intern

Bangalore, India  
May 2013 – July 2013

- Adult content detection for Amazon Android AppStore, leading to a 30% load reduction for human reviewers
- Developed a system to cluster Amazon seller FAQs using Latent Dirichlet Allocation (LDA) and map customer queries to the clusters with logistic regression, SVMs, and supervised LDA

## AWARDS AND HONORS

---

- Project Acyut: A team of 5 undergraduate students that built India's first indigenously developed autonomous humanoid robot. I designed and implemented all aspects of the computer vision and robot behavior modules [\[press coverage\]](#)
  - 3rd Prize at RoboCup 2011 in Autonomous Humanoid Team Soccer
  - 2 silver medals at RoboGames 2011, the (then) world's biggest open robotics event
- Placed 4th in ACM ICPC, Asia Regionals, Kharagpur site in December 2013
- Placed 2nd in Amazon Machine Learning HackEnd 2013 Kindle summary relevance challenge
- Winner of LinkedIn Bangalore HackDay 2015 for developing realtime viewer statistics aggregating system using NodeJS

## PUBLICATIONS

---

- [1] "Interleaved Reasoning for Large Language Models via Reinforcement Learning" [\[pdf\]](#)  
R. Xie, D. Qiu, **D. Gopinath**, D. Lin, Y. Sun, C. Wang, S. Potdar, B. Dhingra  
*Neurips (under submission)*, 2025.
- [2] "Apple Intelligence Foundation Language Models" [\[pdf\]](#)  
T Gunter, Z Wang, C Wang, R Pang, ..., **D. Gopinath**, Apple Inc.  
*arXiv preprint arXiv:2407.21075*, 2024.
- [3] "Simulation and Retargeting of Complex Multi-Character Interactions" [\[pdf\]](#)  
Y Zhang, **D. Gopinath**, Y Ye, J Hodgins, G Turk, J Won  
*ACM Special Interest Group on Computer Graphics (SIGGRAPH)*, 2023.
- [4] "CIRCLE: Capture In Rich Contextual Environments" [\[pdf\]](#)  
JP Araújo, J Li, K Vetrivel, R Agarwal, J Wu, **D. Gopinath**, AW Clegg, K Liu  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [5] "Motion In-betweening for Physically Simulated Characters" [\[pdf\]](#)  
**D. Gopinath**, J. Won, and H. Joo  
*ACM Special Interest Group on Computer Graphics Asia (SIGGRAPH Asia)*, 2022.
- [6] "Physics-based Character Controllers using Conditional VAEs" [\[webpage\]](#)[\[pdf\]](#)  
J. Won, **D. Gopinath**, and J. Hodgins  
*ACM Special Interest Group on Computer Graphics (SIGGRAPH)*, 2022.

- [7] “Transformer Inertial Poser: Attention-based Real-time Human Motion Reconstruction from Sparse IMUs” [\[pdf\]](#)  
Y. Jiang, Y. Ye, **D. Gopinath**, J. Won, A.W. Winkler, and C.K. Liu  
*ACM Special Interest Group on Computer Graphics Asia (SIGGRAPH Asia)*, 2022.
- [8] “Leveraging Demonstrations with Latent Space Priors”  
J. Gehring, **D. Gopinath**, J. Won, G. Synnaeve, A. Krause, and N. Usunier  
*Conference on Neural Information Processing Systems (NeurIPS) (under review)*, 2022.
- [9] “Control Strategies for Physically Simulated Characters Performing Two-player Competitive Sports”.  
[\[webpage\]](#)[\[pdf\]](#)  
J. Won, **D. Gopinath**, and J. Hodgins  
*ACM Special Interest Group on Computer Graphics (SIGGRAPH)*, 2021.
- [10] “A Scalable Approach to Control Diverse Behaviors for Physically Simulated Characters” [\[webpage\]](#)[\[pdf\]](#)  
J. Won, **D. Gopinath**, and J. Hodgins  
*ACM Special Interest Group on Computer Graphics (SIGGRAPH)*, 2020.
- [11] “fairmotion - Tools to load, process and visualize motion capture data” [\[code\]](#)  
**D. Gopinath** and J. Won  
*GitHub, facebookresearch/fairmotion*, 2020.
- [12] “Harnessing Indirect Training Data for End-to-End Automatic Speech Translation: Tricks of the Trade” [\[pdf\]](#)  
J. Pino, L. Puzon, J. Gu, X. Ma, A. D. McCarthy, and **D. Gopinath**  
*International Conference on Spoken Language Translation (IWSLT)*, 2019.
- [13] “Deep Multimodal Fusion for Persuasiveness Prediction” [\[pdf\]](#)  
B. Nojavanasghari\*, **D. Gopinath**\*, J. Koushik\*, T. Baltrušaitis, and L. P. Morency  
*ACM International Conference on Multimodal Interaction (ICMI)*, 2016.

## GRADUATE COURSES

---

|                              |                               |                               |
|------------------------------|-------------------------------|-------------------------------|
| 10701 Machine Learning       | 10807 Topics in Deep Learning | 10705 Intermediate Statistics |
| 11777 Advanced Multimodal ML | 11642 Search Engines          | 11641 ML for Text Mining      |
| 10605 ML for Large Datasets  | 11611 NLP                     | 11513 Intro to Comp Systems   |